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Criteria	Score	
Data Cleaning and Preparation		
Data Analysis and Insight		
GitHub Portfolio Quality		
Total		

PART 1: DATA ANALYSIS AND INSIGHT

This study uses Excel to perform Exploratory Data Analysis (EDA) on the Student Performance dataset.

1. Data Overview

The Student Performance dataset includes demographic information and academic performance in math, reading, and writing for 202 rows across 8 columns, each row represents for an individual student.

Data Column Descriptions

- Race_ethnicity (object): Different racial groups
- Parental_level_of_education (object): The educational background of the student's family
- Math_score (int): Performance in math
- Reading_score (int): Performance in reading
- Writing_score (int): Performance in writing skills
- Total_score (int): The sum of math, reading, and writing scores
- Average_score (float): The mean of the three subject scores

This dataset enables the analysis of the relationship between demographic factors such as gender, race/ethnicity, and parental education and student academic performance, providing a representative basis for evaluating determinants of student success.

2. Data Cleaning

I use the Go To command (Ctrl + G) in Excel to identify missing values in the dataset. The parental_level_of_education column contains 3 missing values, while the average_score column has 4 missing values. All remaining columns have no missing values.

Missing values in average_score were recalculated from math, reading, and writing scores. Missing values in parental_level_of_education were classified as “Unknown”.

Additionally, 3 duplicate rows were removed using Excel's Remove Duplicates feature, resulting in a final dataset of 199 observations.

3. Descriptive Statistics and Insights

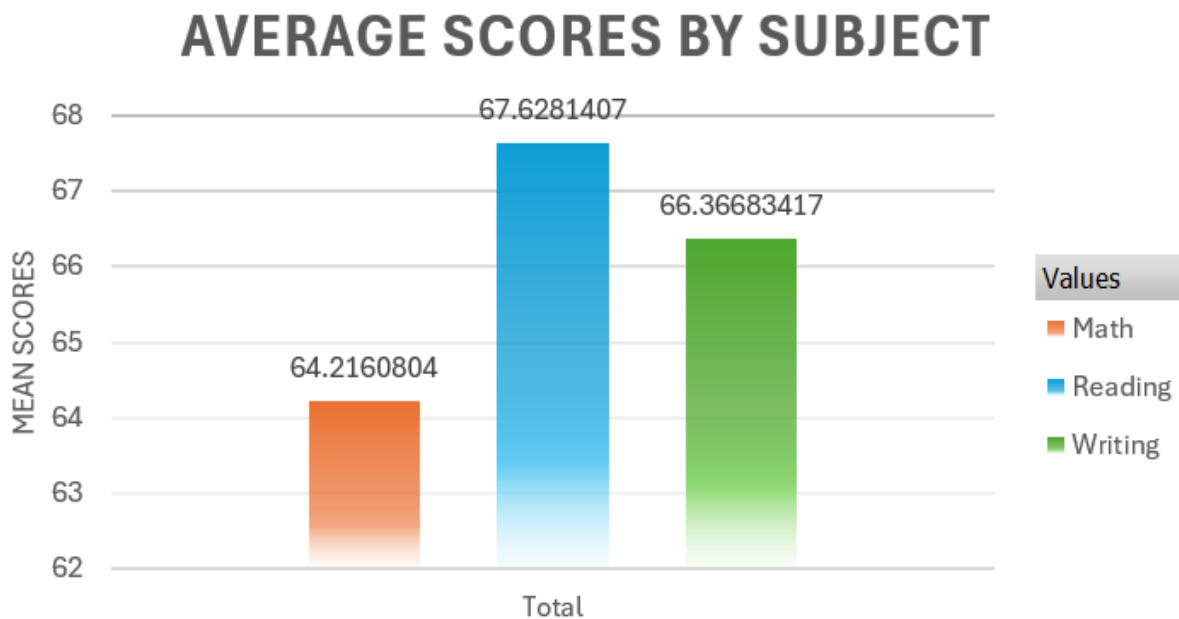
I use the Data Analysis tool in Excel to generate descriptive statistics for students' scores in math, reading, writing, as well as total and average scores.

math_score		reading_score		writing_score		total_score		average_score	
Mean	64.2160804	Mean	67.6281407	Mean	66.3668342	Mean	198.2110553	Mean	66.07035176
Standard Error	1.127361218	Standard Error	1.118896059	Standard Error	1.15259104	Standard Error	3.274682488	Standard Error	1.091560829
Median	65	Median	68	Median	68	Median	199	Median	66.33333333
Mode	58	Mode	67	Mode	74	Mode	151	Mode	50.33333333
Standard Deviation	15.90338706	Standard Deviation	15.78397129	Standard Deviation	16.2592974	Standard Deviation	46.19508127	Standard Deviation	15.39836042
Sample Variance	252.9177199	Sample Variance	249.1337496	Sample Variance	264.364753	Sample Variance	2133.985534	Sample Variance	237.1095037
Kurtosis	0.946460599	Kurtosis	-0.14052162	Kurtosis	0.07261379	Kurtosis	0.341399697	Kurtosis	0.341399697
Skewness	-0.36683033	Skewness	-0.23143742	Skewness	-0.33337789	Skewness	-0.34874371	Skewness	-0.34874371
Range	100	Range	83	Range	90	Range	272	Range	90.66666667
Minimum	0	Minimum	17	Minimum	10	Minimum	27	Minimum	9
Maximum	100	Maximum	100	Maximum	100	Maximum	299	Maximum	99.66666667
Sum	12779	Sum	13458	Sum	13207	Sum	39444	Sum	13148
Count	199	Count	199	Count	199	Count	199	Count	199

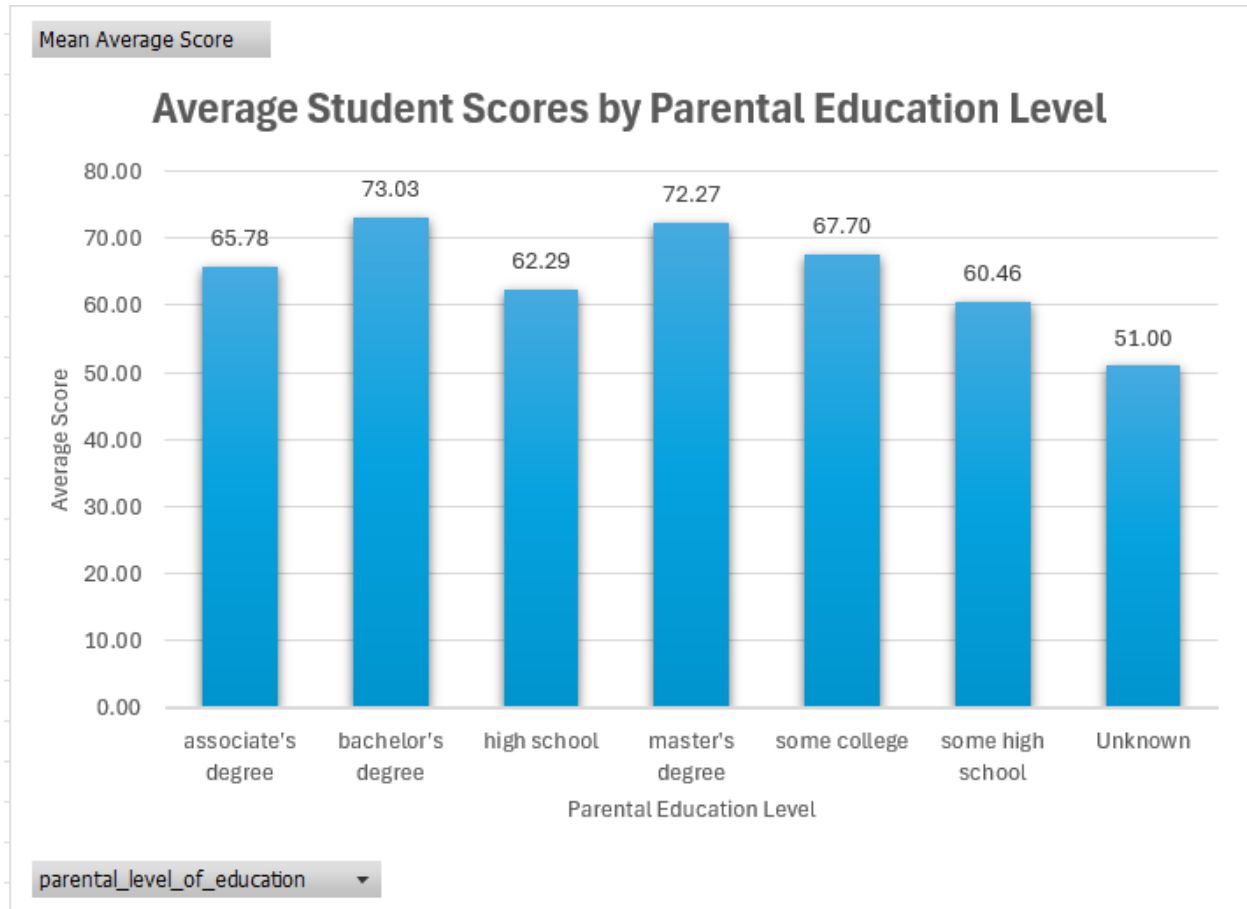
From the results, students scored an average of 64.22 in math, 67.63 in reading, and 66.37 in writing. This shows that students performed slightly better in reading than in the other subjects. The highest score is 100, while the lowest score is quite low, indicating a wide range of results. In addition, the standard deviation of around 15 to 16 suggests there is noticeable variation in student performance.

Key Insight:

Math Reading Writing



The descriptive statistics show that students have higher average scores in reading and writing than in mathematics. The lower mean score in mathematics suggests that this subject may be more challenging for students. This result indicates a possible gap between students' math skills and their language-based skills.



The descriptive statistics show that students whose parents have higher levels of education tend to achieve higher average scores. The mean scores increase gradually as parental education level rises. This suggests a possible relationship between parental educational background and students' academic performance.